USACE ERDC CONSTRUCTION & ENGINEERING RESEARCH LABORATORY (CERL) COMMERCIAL SOLUTIONS OPENING (CSO 2026) - EXECUTIVE COVER LETTER

Date: October 30, 2026

Primary Offeror: Humphrey Virtual Farms LLC (701 E Drew St., Madill, OK 73446)

Business Status: Non-Traditional Defense Contractor (NTDC), Small Disadvantaged Business (SDB), Minority-Owned Small Business

SAM.gov UEI: S1M4ENLHTDH5 | CAGE Code: 1AHA8 | NAICS: 541715, 541511, 541330

Authorized Representative: Jeffery Humphrey, Founder, President & CEO | (572) 358-9978 | humphreyvirtualfarm@gmail.com

Proposed Solution Title: Project Ebony v8.0.0: Air-Gapped Tri-Brain OT/SCADA Safety Interlock, Automated Bridge/Pavement NDE & WORM Merkle Provenance Testbed

CERL Areas of Interest: Operational Energy; Sustainment Management System; Robotics for Engineering Operations

Cross-Referenced Federal Filings: Active DoD CDAO Tradewinds Solutions Marketplace Docket 10-26-4046 (Status: Received | Video SHA-256: c597865785eae6fdad6fdf6803a2710aaa0fb5c2fa28fe512b6ed76ef8b7a7fd | Merkle Root: bb39eeaa66a39ad2e0c0ab74ba4a5c4faac2bf0d98babc0362482b97607151a1) and USACE ERDC BAA Filings (CERL, ITL-4, GSL-19).

1. Executive Purpose & Statutory Eligibility (10 U.S.C. 4022 / 10 U.S.C. 3458)

Humphrey Virtual Farms LLC respectfully submits this complete four-part Commercial Solutions Opening (CSO) package to the U.S. Army Engineer Research and Development Center (ERDC) Construction and Engineering Research Laboratory (CERL). As a verified U.S. Non-Traditional Defense Contractor (NTDC) and Small Disadvantaged Business (SDB) actively registered in SAM.gov (UEI: S1M4ENLHTDH5 | CAGE: 1AHA8), Humphrey Virtual Farms LLC qualifies for streamlined commercial and prototype Other Transaction Authority (OTA) or FAR/DFARS award with 0% required cost sharing.

2. Summary of Operational Impact for CERL Installations & Facilities

Per U.S. GAO reports GAO-21-81 and GAO-24-106137 and Joint CISA/DoD Volt Typhoon advisories (AA23-144A / AA24-038A), DoD’s 500+ installations, 300,000+ buildings, and $4.2B+ in operational energy and utility assets rely on commercial PLCs scanning at 50 to 200ms, whereas arc-flash faults destroy switchgear in <4.16ms and thermal excursions occur in 6.0ms. Project Ebony v8.0.0 provides a live TRL 4 air-gapped software emulation testbed on local loopback 127.0.0.1 (0% cloud dependency, 0 external retrieval APIs, 637 C2 modules, 85/85 passing unit test suites, 52 SQLite WORM tables) modeling a

<50us safety loop and 36.75ns SiC MOSFET cutoff target (Brain 1), isolated ASTM D6433 structural/pavement NDE diagnostics cutting $4,600/day inspections to $250/sortie (Brain 2), and SHA-256 WORM Merkle provenance (Brain 3).

3. Non-Proprietary & Unclassified Compliance Statement

In strict compliance with ERDCWERX CSO instructions, this submission contains no classified, confidential, or restricted proprietary markings and may be shared across ERDC teams. Background commercial software modules developed at private expense are governed by standard DFARS 252.227-7018 / FAR 52.227-14 provisions upon award. Respectfully submitted by Jeffery Humphrey, Founder, President & CEO, Humphrey Virtual Farms LLC.